SmartPOS inventory guide

Every inventory screen: what it does, why it exists, and what is still missing

How inventory works

Inventory answers three questions every owner asks: **what do I have**, **where did it go**, and **what is missing**. SmartPOS keeps two kinds of stock. **Products** are sold as they are, like a bottle of Fanta. **Ingredients** are never sold on their own; they are used to make food, like rice, eggs and cooking oil. Every change to stock is recorded with a reason, so the numbers can always be explained.

The life of an ingredient

Add it once, with its units and cost → Restock when a delivery arrives → Recipes take it out when food is paid for → Staff record anything wasted

Count the storeroom regularly → Usage and losses shows what is missing → Low stock and expiry warn you early → Restock again

Reason	When it is recorded	Example
Sale	An order is paid. Products drop by the amount sold; ingredients drop by what their recipes use.	10 plates of rice take 1.5 Kg of rice
Restock	A delivery is entered.	3 trays of eggs add 90 pieces
Waste	Staff record stock that was lost, with a reason.	2 Kg of tomatoes spoiled
Stock count	Someone counts and enters what is really there.	Expected 20 Kg of beef, counted 18 Kg
Opening stock	A new item is added with stock already on the shelf.	A new ingredient starts with 50 Kg

Status: Everything in this guide has been built and type-checked, but not yet tested with real sales. The app currently talks to the backend on Render, which does not have this work yet. Test it against your local backend first, then deploy it, before shops rely on it.

Who can do what

Permission	Lets a worker
inventory.view	See Stock Levels, Ingredients, Low Stock and Expiry, Usage and Losses, and stock history.
inventory.adjust	Restock, record waste and change counted stock. Comes with inventory.view.
products.manage	Add and edit products and ingredients, recipes and pictures. Saving a Stock Count also needs it.
reports.view	See Activity Logs: who changed which product, and what changed.

Owners can do everything. The Stock keeper role preset gives all three inventory permissions.

The inventory screens

In the sidebar under Inventory, in this order: Stock Levels, Ingredients, Usage and Losses, Stock Count, Orders, Categories, Products, Low Stock and Expiry, Activity Logs.

Stock Levels

Inventory > Stock Levels

What it is. One list of everything whose stock is counted, emptiest first, with how much is left and a Restock button.

Why we built it. An owner needs one place to see the whole shop at a glance, not a page per product. Sorting emptiest first puts what needs action at the top.

How it works

Part	What it does
Counted, Low, Out	How many items are tracked, at their alert level, or at zero or below.
Each row	Name, alert level and stock left. Red when low or out. Tap the name to open its history.
Restock	Opens the delivery window. Only for workers with inventory.adjust.

> **Real example:** A bar opens the page on Saturday morning and sees Primus at 6 bottles in red. It taps Restock and enters the crates that just arrived.

Still missing

- Sorting by value, so the most expensive stock is easy to watch.
- Separate stock for each branch, if a company has several shops.

Ingredients

Inventory > Ingredients

What it is. The table of everything food is made from: stock, alert level, cost, stock value and expiry, with filters and actions on each row.

Why we built it. Ingredients are managed differently from products you sell: they are bought in bulk, used in small amounts and never appear on the till. Keeping them on their own page stops the Products catalog becoming cluttered, and gives owners one place to control the kitchen's stock.

How it works

Part	What it does
Totals	Number of ingredients, total stock value, how many are low or out, and how many expire soon.
Filters	All, Low, Out, Expiring in 7 days, and **Needs setup**: ingredients with no cost or no alert level, listed in red with the reason.
Stock shown	In buying units where that helps: 2 Tray + 20 Piece rather than 80 Piece.
Row actions	Restock, Record waste, History, Edit, and Deactivate.

> **Real example:** The owner taps Needs setup and sees "Salt: no cost, so dish costs are too low". Adding the price fixes the cost of every dish that uses salt.

Still missing

- Importing many ingredients at once from a spreadsheet.
- Supplier and barcode, which were removed from this page for now.
- Printing a stock list for counting on paper.

Add and edit an ingredient

Ingredients > Add ingredient, or Edit on a row

What it is. A full page that walks through everything an ingredient needs, with a live summary on the side.

Why we built it. Getting an ingredient right once makes every later step correct: deliveries, recipes, costs and alerts. The page asks plain questions in a fixed order so nothing is forgotten.

How it works

Part	What it does
Picture	Search free pictures by name, paste an image link copied from any website, or upload from the device. Optional.
About	Name, with a warning if it already exists, and storage notes such as "keep frozen".
How the kitchen uses it	Required, with no default, so eggs can't end up counted in Kg by accident. See the next section.
How you buy it	Optional: the bigger unit it arrives in, how many units are inside, and its price. The cost per unit is worked out for you.
Stock	What is there now and when to warn. Both in the kitchen unit.
Summary and Used in	Status, stock value and cost, and the dishes whose recipes use it.
Save and add another	Saves and clears the page for the next ingredient.

> **Real example:** Eggs: kitchen uses Piece, bought by the Tray, 30 in one, 3,000 RWF a tray. The page shows "1 Tray = 30 Piece" and a cost of 100 RWF per egg.

Still missing

- Supplier, expiry date, barcode and SKU, removed from this page for now and still stored for existing ingredients.
- Yield: how much is usable after peeling or deboning. Work around it by writing recipes in what is taken from storage.

Units of measure

Why units are designed this way. Professional systems keep one base unit per ingredient and link bigger buying sizes to it. SmartPOS does the same. Weight, liquid and counted items cannot be converted into each other without measuring, so cups and spoons are offered for liquids only. Local measures such as sacks, heaps and basins vary widely between sellers, so each shop enters its own size rather than the app guessing one.

How the kitchen uses it

Unit	Meaning	Use it for	Recipe can be typed in
Kg	Weighed	Rice, sugar, beans, beef	g or Kg
Gram	Very small weights	Yeast, saffron	g or Kg
Litre	Poured	Oil, milk, sauce	ml, Tsp (5 ml), Tbsp (15 ml), Cup (250 ml), Litre
Millilitre	Very small amounts poured	Vanilla, food colour	The same as Litre
Piece	Counted one by one	Eggs, onions, whole chicken	Tap ¼, ½, 1 or 2, or type
Bunch	Tied together	Coriander, spring onions	Tap ¼, ½, 1 or 2, or type
Bottle	Used by the bottle	Soda, water, beer	Tap ¼, ½, 1 or 2, or type
RWF	Bought by price	Spices, garlic, ginger	The amount of money used
Something else	Your own word	Anything unusual	Your own unit

How you buy it

Kg, Litre, Sack, Bag, Basin, Bucket, Tray, Crate, Carton, Pack, Jerrycan and Dozen. Dozen fills in 12 by itself; every other size is entered by the shop.

Worked examples

Item	Kitchen uses	Bought by	Inside one	Recipe
Eggs	Piece	Tray	30	2 per plate of rice
Maggi cubes	Piece	Pack	100	1 cube, or ½
Onions	Piece	Kg	About 8 (weigh and count once)	1 onion, or ½
Whole chicken	Piece	(as it arrives)		½ for Half chicken, 1 for Full
Rice	Kg	Sack	25	150 g
Cooking oil	Litre	Jerrycan	20	2 Tbsp
Spices	RWF	(by price)		5 RWF per plate

> **Two rules for recipes:** Recipes are per plate sold, not per pot. If a pot of 20 plates uses 100 RWF of spices, enter 5. And enter what is taken from storage, not what ends up on the plate: 200 g of raw beef, not 150 g of cooked meat.

Still missing

- Converting an existing ingredient's unit. Changing Gram to Kg does not convert its stock or recipes; both must be updated together.
- More than one buying size for the same ingredient, such as eggs by the tray and by the carton.
- Cups and spoons for dry goods, which need a measured weight per ingredient.

Recipes

On any product with Show in kitchen turned on

What it is. What one serving of a dish is made from. Each paid sale takes those amounts out of ingredient stock automatically.

Why we built it. Without recipes the app knows a burger was sold but not that a bun, beef and onions left the storeroom. Recipes turn sales into ingredient use, and show what each dish really costs.

How it works

Part	What it does
Add ingredients	Tap an ingredient's name to add it, then enter the amount.
Measures	Tap g or Kg for weighed items; ml, Tsp, Tbsp, Cup or Litre for liquids; ¼, ½, 1 or 2 for counted items.
Check line	Shows what will leave stock, such as "1 Tbsp = 15 ml · Takes 0.03 Litre from stock".
Cost to make one	The ingredients' cost at today's prices, and the profit at the selling price.
On the till	If the ingredients left can't make the amount in the cart, it warns "Ingredients for 3 left". The sale can still go ahead.

> **Real example:** Rice plate: Rice 150 g, Eggs 2, Cooking oil 1 Tbsp, Spices 5 RWF. Cost to make one is shown with the profit at 2,500 RWF. Selling 10 plates takes 1.5 Kg of rice and 20 eggs out of stock.

Still missing

- A log of who changed a recipe and when.
- Recipes inside recipes, such as a batch of sauce used by several dishes.
- Saving the cost of each sale; this is being added on the backend as unit cost.

Restock

The Restock button on Stock Levels, Ingredients, Low Stock and history pages

What it is. Records a delivery: how much arrived, optionally what was paid, a new expiry date and a note.

Why we built it. Stock only stays right if deliveries are entered. Entering what was paid also keeps the cost of every dish up to date when prices change.

How it works

Part	What it does
By Tray / By Piece	For items bought in bigger units: enter 3 trays and the app adds 90 pieces, showing "Adds 90 Piece to stock".
Total paid	Optional. Works out the new cost per unit, which recipes then use.
New expiry date	Optional. Replaces the ingredient's expiry date.
Note	Saved in the history with the delivery, such as "3 Tray · From Kimironko market".

> **Real example:** Eggs went up to 3,600 RWF a tray. Restock 3 trays with 10,800 RWF paid: stock rises by 90 eggs and the cost becomes 120 RWF per egg, so the rice plate's cost rises by 40 RWF.

Still missing

- Supplier records and purchase orders: who you buy from, at what price, over time.
- Average cost. The newest delivery's price replaces the old one.

- Separate batches with their own expiry dates.
- A log of price and expiry changes made at restock.

Record waste

The waste button on each ingredient row

What it is. Takes lost stock out with a reason: Spoiled, Expired, Spilled, Burnt or Damaged.

Why we built it. Without it, waste looks the same as theft or a counting mistake. Recording it separately shows the owner how much is thrown away, and keeps the unexplained loss honest.

How it works

Part	What it does
Amount and reason	The server refuses more than is in stock, so a typing mistake can't empty it.
Alerts	If waste takes an ingredient to its alert level, owners are told.

> **Real example:** On Monday 2 Kg of tomatoes went soft. The cook records 2 Kg, Spoiled. Usage and Losses later shows tomato waste of 8% for the month.

Still missing

- A photo as proof, or a manager's approval for large amounts.
- A waste report by reason and by worker.

Stock Count

Inventory > Stock Count

What it is. Walk the storeroom, type what is really there, and see the difference from what the app expected, with its value, before saving.

Why we built it. Every system drifts from reality. A count is the only way to know the true stock, and it is what makes losses visible.

How it works

Part	What it does
Choose	Ingredients only, or everything counted.
Each row	Expected amount, the count typed, and the difference in red when stock is missing.
Total	How many differences, and what they are worth at today's cost.
Save count	Saves the differences only, then opens Usage and Losses.

> **Real example:** Every Sunday night the manager counts. Beef was expected at 12 Kg but there are 10 Kg: the row shows -2 Kg in red, worth 16,000 RWF.

Still missing

- Workers with only inventory.adjust can open the page but saving fails, because the server also asks for products.manage.
- Blind counts, where the counter doesn't see the expected amount first. These give more honest counts.
- If one row fails to save the others still save; the app names the ones that failed.

Usage and Losses

Inventory > Usage & Losses, and after every Stock Count

What it is. Compares what recipes used and what was recorded as waste with what a stock count actually found, for 7, 30 or 90 days.

Why we built it. This is the owner's control panel for the kitchen. A count alone says stock is missing; this report says how much nothing explains, and puts the biggest money loss at the top.

How it works

Part	What it does
Used in food	What recipes took out, valued at today's cost.
Wasted	What staff recorded, and the waste rate.
Unexplained loss	Stock a count found missing that no sale or recorded waste explains.
Each ingredient	Used, wasted, received, count difference, missing value, and the missing share of use. 5% or more is marked Worth checking.

> **Real example:** Recipes used 10 Kg of beef, 0.5 Kg of waste was recorded, and the count found 2 Kg missing. The report shows 19% and marks it Worth checking. The likely causes are waste nobody recorded, a recipe amount set too low, or theft.

Still missing

- Losses by worker or shift, to see whether a problem follows one person.
- Values at the price actually paid. It uses today's cost.
- Trends over time, such as a chart of waste each week.

Low Stock and Expiry

Inventory > Low Stock & Expiry

What it is. What has reached its alert level, and what expires within 7, 30 or 90 days, including anything already expired.

Why we built it. It is cheaper to reorder before running out and to use stock before it expires. Owners also get a live notification the moment a sale takes something to its alert level.

> **Real example:** Milk reaches its alert of 5 Litres at lunchtime. The owner's phone shows "Milk is low on stock, 4.5 Litre left" and the shop reorders before dinner.

Still missing

- Reorder suggestions based on how fast each item is used.
- Sending an order straight to the supplier.

History and Activity Logs

History on any item; Inventory > Activity Logs

What it is. Stock history shows every change to one item: sale, restock, count or waste, by whom, when, and how much was left. Activity Logs show who added or edited products and exactly what changed.

Why we built it. When a number looks wrong, the owner can see why without asking anyone. It also discourages mistakes and theft, because every change has a name on it.

> **Real example:** Rice dropped faster than expected. Its history shows "Used in 40 × Rice plate" on Saturday and a waste of 3 Kg recorded by the cook, which explains the drop.

Still missing

- Recipe changes, and price or expiry changes made at restock, are not in Activity Logs yet.

Inventory on the till and in the kitchen

- **Stock drops when the order is paid**, not at checkout, so an order that is never paid never touches stock.
- **The cart warns, it never blocks**: "5 Bottle in stock" or "Ingredients for 3 left". Stock can go below zero, because a shop's count is often slightly off and refusing would stop a real sale.
- **Ingredients never appear on the till**, and the server refuses to sell them.
- **The kitchen only sees products with Show in kitchen turned on**, and those are the products recipes are set on.

What is still missing

In the order I would do them. The first two matter before any shop relies on inventory.

#	What	Why it matters	Real example
1	**Test, then deploy** the backend work, and run its migrations	The app currently talks to an older backend without this work.	Image search returned "productId must be a UUID" because Render doesn't have the route.
2	**Kitchen progress apart from payment**	Orders paid at checkout skip the kitchen board, so no ready alert goes out.	A takeaway paid upfront is never shown to the cook.
3	**Supplier and purchase records**	Shows who you buy from, what you paid over time, and lets you compare suppliers.	Tomatoes from one supplier cost 15% more for the same basin.
4	**Yield**, the usable share after peeling or deboning	Recipes can then be written in cooked or plate amounts.	1 Kg of beef gives 750 g of meat after bone and fat.
5	**Average cost** instead of the latest price	Profit reports stay steady when one delivery is unusually cheap or expensive.	One cheap sack of rice shouldn't make every plate look more profitable.
6	**Batches with their own expiry**	Older stock is used first and expiry is accurate.	Two deliveries of milk with different dates.
7	**Recipes inside recipes**	Batch items like sauce or dough used by several dishes.	One pot of sauce serves burgers and pasta.
8	**Logs for recipe and price changes**	Owners see who changed what affects cost.	A recipe's beef was lowered from 200 g to 120 g.
9	**Losses by worker and shift**	Points to where a problem happens.	Losses appear only on the night shift.
10	**Refunds that return stock**	A cancelled paid order should put ingredients back.	A plate sent back unopened.
11	**Several branches**	Each shop needs its own stock.	A restaurant with two locations.
12	**Selling part of a unit on the till**	The till sells whole numbers only.	Selling 1.5 Kg of meat by weight.

Habits that make inventory work

- **Enter every delivery** the day it arrives, with what was paid.
- **Record waste** when it happens, not at the end of the week.
- **Count weekly**, by the same person, at the same time, before deliveries arrive.
- **Check Needs setup** until no ingredient is missing a cost or alert level.
- **Look at Usage and Losses after each count**, starting from the top row.